

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 What is the length of the vector projection of $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ onto $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$?
- A. $\frac{3\sqrt{13}}{13}$ units
- B. $\frac{3\sqrt{10}}{10}$ units
- C. $\frac{3\sqrt{10}}{13}$ units
- D. $\frac{3\sqrt{13}}{10}$ units
- 2 Which of the following statement does *not* describe a Bernoulli random variable?
- A. Selecting a faulty phone in a random quality control check.
- B. Guessing the correct answer to a true/false question.
- C. The number of tails when tossing 2 coins.
- D. A missile hitting a target.
- 3 The polynomial $P(x) = x^4 + ax^3 - 3x^2 + bx - 2 = 0$ has roots -1 and 2 , one of which is a triple root. Find the values of a and b .
- A. $a = -1$ and $b = 2$
- B. $a = 2$ and $b = -5$
- C. $a = 1$ and $b = 3$
- D. $a = 1$ and $b = -5$

4 What is the derivative of $\tan^{-1} 4x$?

A. $\frac{4}{\sqrt{16 - x^2}}$

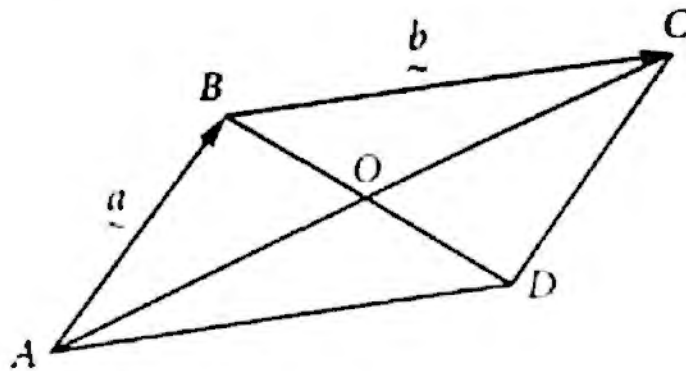
B. $\frac{4}{\sqrt{x^2 - 16}}$

C. $\frac{4}{1 - 16x^2}$

D. $\frac{4}{1 + 16x^2}$

5 In the parallelogram ABCD shown, the point of intersection of the diagonals is at O, where O is the midpoint of both $\overrightarrow{AC}$ and $\overrightarrow{BD}$.

The vector $\overrightarrow{OC}$ is equal to



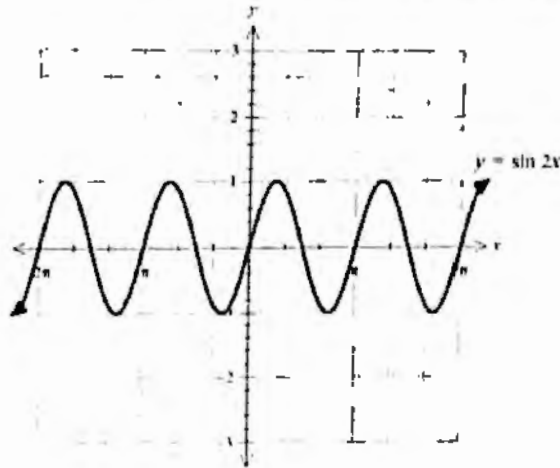
A. $\frac{1}{2}(\overrightarrow{AB} - \overrightarrow{BC})$

B. $\frac{1}{2}(\overrightarrow{AB} + \overrightarrow{BC})$

C. $(\frac{1}{2}\overrightarrow{AB} - \overrightarrow{BC})$

D. $\frac{1}{2}(\overrightarrow{BC} - \overrightarrow{AB})$

- 6 The function $y = \sin(2x)$ is shown in the diagram below.



If this function is transformed using steps I, II and III as below:

I : Reflected in x -axis

II: Vertically translated 1 unit down

III: Dilated horizontally by a scale factor of 2.

Which one of the following equations would represent the transformed function?

- A. $y = -(\sin x + 1)$
- B. $y = 2(\sin(2x) - 1)$
- C. $y = -(\sin(4x)) + 2$
- D. $y = -2 \sin(2x) - 1$

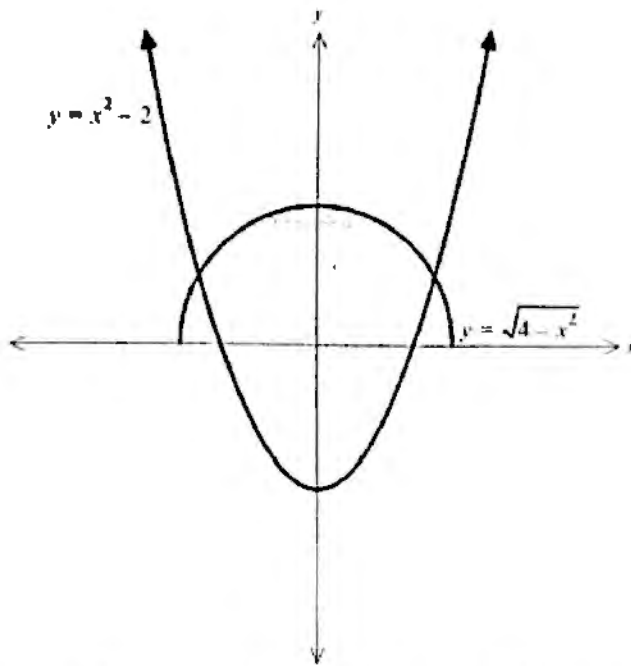
- 7 The continuous random variable, X , has the following probability density function:

$$f(x) = \begin{cases} ax^2(4-x), & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

What could be the value of a ?

- A. $\frac{81}{4}$
- B. $\frac{4}{81}$
- C. 0
- D. 1

- 8 The area enclosed by the curves $y = \sqrt{4 - x^2}$ and $y = x^2 - 2$ is rotated about x -axis.



Which one of the expressions could be used to calculate the volume?

- A.
$$V = \pi \int_0^{\sqrt{3}} \left(\sqrt{4 - x^2} - (x^2 - 2) \right)^2 dx$$
- B.
$$V = \pi \int_0^{\sqrt{3}} (3x^2 - x^4) dx$$
- C.
$$V = 2\pi \int_0^{\sqrt{3}} (3x^2 - x^4) dx$$
- D.
$$V = 2\pi \int_0^{\sqrt{3}} \left(\sqrt{4 - x^2} - (x^2 - 2) \right)^2 dx$$

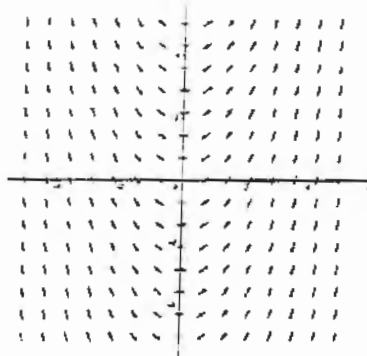
- 9 A large tank initially holds 2500L of water in which 100kg of salt is dissolved. A solution containing 4kg of salt per litre flows into the tank at a rate of 10L per minute. The mixture is stirred continuously and flows out of the tank through a hole at a rate of 14L per minute.

A differential equation for Q , the number of kilograms of salt in the tank after t minutes, is given by

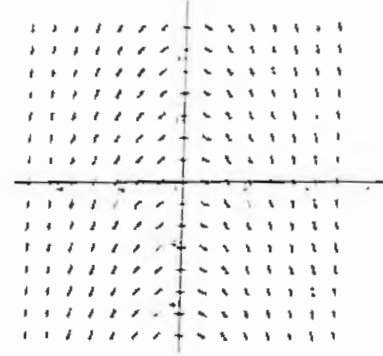
- A. $\frac{dQ}{dt} = 40 - \frac{7Q}{2(625 - t)}$
 B. $\frac{dQ}{dt} = 40 + \frac{7Q}{2(625 - t)}$
 C. $\frac{dQ}{dt} = 40 - \frac{7Q}{2(625 + t)}$
 D. $\frac{dQ}{dt} = 40 + \frac{7Q}{2(625 + t)}$

- 10 Which of the following slope field best represents the differential equation $y' = x + y$

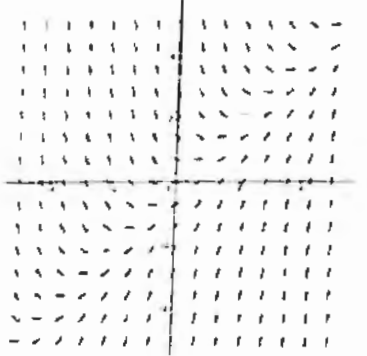
A.



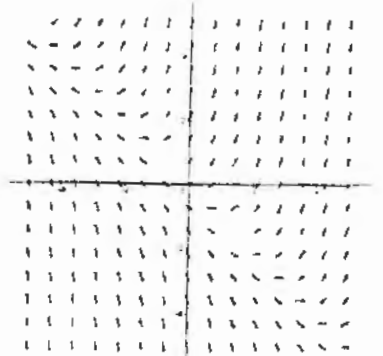
B.



C.



D.



END OF SECTION I

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question on a new writing sheet. Extra writing sheets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

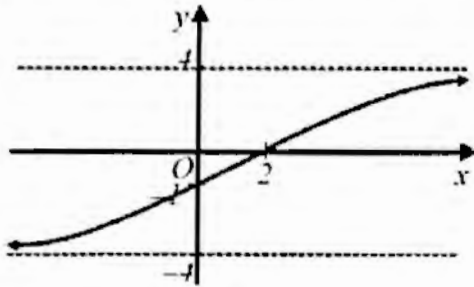
- (a) Evaluate $\int_{-1}^0 2x\sqrt{1+x} \, dx$ using the substitution $u = 1+x$. 3
- (b) Find the coefficient of x^4 in the expansion of $(2x^2 - \frac{1}{x})^{11}$. 3
- (c) (i) Solve the differential equation $\frac{dy}{dx} = \frac{x^2}{y^2}$ 1
- (ii) Find the solution of this equation that satisfies the initial condition $y(0) = 2$. 1
- (d) Consider the function $y = \cos^{-1}(\sin x)$.
- (i) Find $\frac{dy}{dx}$. 1
- (ii) Clearly explain why $\frac{dy}{dx}$ can be written as 1 or -1 . 2
- (iii) State the domain and range for this function. 2
- (iv) Hence, neatly sketch the function over the domain $0 \leq x \leq 2\pi$. 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Express $\sqrt{3} \cos x - \sin x$ in the form $A \cos(x + \alpha)$ where α is an acute angle. 2
- (ii) Hence, or otherwise, solve the equation $\sqrt{3} \cos x - \sin x = 1$ for $0 \leq x \leq 2\pi$ 2

- (b) The graph of $y = f(x)$ is illustrated. The lines $y = \pm 4$ are horizontal asymptotes.



Without using Calculus, sketch each of the graphs below. In each case, clearly label any maxima or minima, intercepts and the equations of any asymptotes.

- (i) $y = f(x + 2)$ 1
- (ii) $y = |f(x)|$ 1
- (iii) $y = \sqrt{f(x)}$ 2
- (c) Find the values of k for which the non-zero vectors $\vec{a} = k^2\vec{i} + 2\vec{j}$ and $\vec{b} = 3\vec{i} - (2 + 2k)\vec{j}$ are perpendicular. 3
- (d) Use the Principle of Mathematical Induction to show that if x is a positive integer then $(1 + x)^n - 1$ is divisible by x for all positive integers $n \geq 1$. 4

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $4x^3 - 27x + k = 0$ has a double root. Find the possible value(s) of k . 3
- (b) Use vectors to prove that the sum of the squares of the lengths of the two diagonals of a parallelogram is equal to the sum of the squares of the lengths of the four sides. 3
- (c) A company manufactures lithium batteries using a mixture of digital and traditional techniques. Data shows the probability that a random case will fail quality control is 8%. An inspector selects a random batch of 50 cases from the warehouse.
- Let X be the binomial random variable of the number of cases that do not pass the inspection.
- (i) What is the mean, variance and standard deviation for this distribution, correct to 2 decimal places? 3
- (ii) Find the probability that the number of cases that fail to pass within one standard deviation of the mean. 2
- (iii) A new company standard insists that the number of failures in a batch must be no more than one standard deviation above the mean. Batches that fail to meet this standard are rejected. 2
- What is the probability of a batch being rejected?
- (iv) Due to the new regulations and the number of rejected batches, the company improves its manufacturing process so that the new experimental probability of failure is reduced to 4%. Find the new probability that a batch will be accepted. 2

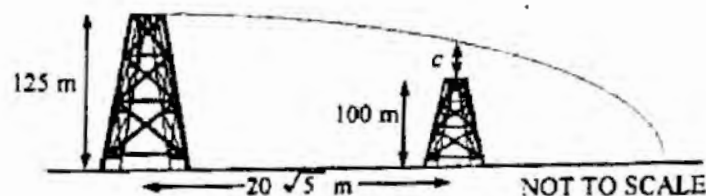
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Express $\sin 2\theta$ and $\cos 2\theta$ in terms of $t = \tan \theta$. 2
- (ii) Hence, or otherwise, prove that, for $0 < \theta < \frac{\pi}{2}$, 2

$$\frac{1 + \sin 2\theta - \cos 2\theta}{1 + \sin 2\theta + \cos 2\theta} = \tan \theta$$

- (b) A projectile is shot horizontally from the top of a 125 metre tower with a velocity of V metres per second. The projectile clears a building of height 100 metres by a distance of c metres, as shown in the diagram below. The two towers are $20\sqrt{5}$ metres apart. Air resistance is ignored and use $g = 10 \text{ m/s}^2$.



- (i) Derive the six equations of motion. 2
- (ii) Show that the displacement vector of the projectile at any time t is given by 1
 $s(t) = Vt \mathbf{i} + (125 - 5t^2) \mathbf{j}$.
- (ii) Show that $V = \frac{100}{\sqrt{25-c}}$. 2
- (iii) Prove that the minimum speed of projection for the projectile to just clear the 100 m tower is 20 m/s 1
- (iv) Hence, find how far past the 100 metre tower the projectile will strike the ground. 2
- (v) Determine the velocity and the angle of projection when it strikes the ground. 3

End of Question 14

End of paper

Ext 1 Solutions

Monday, 31 August 2020 8:39 AM

$$\begin{aligned}
 1. \quad | \text{Proj}_{\underline{b}} \underline{a} | &= \left| \frac{\underline{a} \cdot \underline{b}}{\underline{b} \cdot \underline{b}} \left(\underline{b} \right) \right| \\
 &= \frac{6-3}{4+9} \begin{pmatrix} 2 \\ -3 \end{pmatrix} \\
 &= \frac{3}{13} | \sqrt{13} | \\
 &= \frac{3\sqrt{13}}{13} \quad \boxed{A}
 \end{aligned}$$

2. $\boxed{C}$

$$\begin{aligned}
 3. \quad &\alpha, \alpha, \alpha, \beta \\
 &P(\alpha) = P'(\alpha) = P''(\alpha) = 0 \\
 &P'(x) = 4x^3 + 3ax^2 - 6x + b \\
 &P''(x) = 12x^2 + 6ax - 6 \\
 &2x^2 + ax - 1 = 0 \\
 &\quad \underline{-1, -1, -1, 2} \\
 &\quad \cancel{2x^2 + 1 + 1 = 2 - 2 = 0} \\
 &\quad 1 + 1 - 2 + 1 - 2 - 2 = -3 \\
 &\quad -1 - 1 - 1 + 2 = -a \\
 &\quad \boxed{a=1} \\
 &P(-1) = 0, \quad 1 - 1 - 3 - b - 2 = 0 \\
 &\quad \boxed{b=-5} \quad \boxed{D}
 \end{aligned}$$

$$\begin{aligned}
 4. \quad &y = \tan^{-1} 4x \\
 &y' = \frac{1}{1+16x^2} x^4 = \boxed{D}
 \end{aligned}$$

5. $\boxed{B}$

6. $\boxed{A}$

$$7. \int_1^4 (4ax^2 - ax^3) dx = 1$$

$$\left[4a \frac{x^3}{3} - \frac{ax^4}{4} \right]_1^4 = 1$$

$$a \times \left[64 \times \frac{4}{3} - 64 - \frac{4}{3} + \frac{1}{4} \right] = 1$$

$$a \left[\frac{64}{3} + \frac{1}{4} \right] = 1$$

$$a \left[\frac{243}{12} \right] = 1 \Rightarrow a = \frac{12}{243} \quad \boxed{B}$$

$$8. \quad 4 - x^2 = x^4 - 4x^2 + 4$$

$$x^4 - 3x^2 = 0$$

$$x^2(x^2 - 3) = 0, \quad x = \pm\sqrt{3}, 0$$

$$V = 2\pi \int_0^{\sqrt{3}} \left[(4 - x^2) - (x^2 - 2)^2 \right] dx$$

$$= 2\pi \int_0^{\sqrt{3}} (4 - x^2 - x^4 + 4x^2 - 4) dx$$

$$= 2\pi \int_0^{\sqrt{3}} (3x^2 - x^4) dx \quad \boxed{C}$$

$$9. \quad \frac{dQ}{dt} = \frac{4 \text{ kg} \times 10 \text{ L/min} - Q \times 14 \text{ L/min}}{(2500 + (10 \text{ kg} - 14 \text{ L}) \text{ min})}$$

$$= 40 - \frac{14Q}{(2500 - 4)}$$

$$= 40 - \frac{7Q}{2(625 - t)} \quad \boxed{A}$$

10. $\boxed{D}$

Question 11

Question 11

a) $u = 1+x$, $x = (u-1)$
 $du = dx$

$$\begin{aligned} I &= \int_0^1 2(u-1)u^{1/2} du \\ &= \int_0^1 2u^{3/2} - 2u^{1/2} du \\ &= \left[\frac{2u^{5/2}}{5} \times 2 - \frac{2u^{3/2}}{3} \times 2 \right]_0^1 \\ &= -\frac{8}{15} \end{aligned}$$

This question was done well overall.

- Only few forgot 2 at the front

- A very few forgot to integrate and they just sub in the value.

b) $\left(2x^2 - \frac{1}{x}\right)^{11}$

the general term is given by

$$\begin{aligned} {}^{11}C_k (2x^2)^{11-k} \left(-\frac{1}{x}\right)^k \\ = {}^{11}C_k 2^{11-k} x^{22-2k} (-1)^k x^{-k} \end{aligned}$$

for x^4 , $22-3k=4 \Rightarrow \boxed{k=6}$

the coefficient of x^4 ;

$$\begin{aligned} {}^{11}C_6 2^5 (-1)^6 \\ = 14784 \end{aligned}$$

- many students did find the correct value of k .

- some students expanded without using the general term

- some wrote the general term from backwards and got $\boxed{k=5}$

- very few wrote $\boxed{-14784}$ which is marked incorrect.

(c) (i) $\frac{dy}{dx} = \frac{x^2}{y^2}$

$y^2 dy = x^2 dx$ (variable-separable)

integrate both sides,

$\frac{y^3}{3} = \frac{x^3}{3} + C$

$y^3 = x^3 + 3C$

(ii) $y(0) = 2 \Rightarrow \boxed{C = \frac{8}{3}}$

- very well done on this part
- some students wrote $y = x + 2$ instead

d) (i) $y = \cos^{-1}(\sin x)$

$y' = \frac{-\cos x}{\sqrt{1 - \sin^2 x}}$

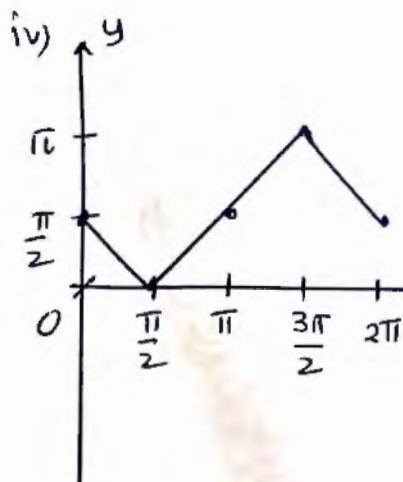
very well done

(ii) $y' = \frac{-\cos x}{|\cos x|} \begin{cases} y' = -1 & \text{when } \cos x > 0 & 1^{\text{st}} \text{ \& } 4^{\text{th}} \text{ Quad.} \\ y' = 1 & \text{when } \cos x < 0 & 2^{\text{nd}} / 3^{\text{rd}} \text{ Quad.} \end{cases}$
 $|\cos x| = \pm 1$

Poor explanation overall

(iii) $D_f: (-\infty, \infty)$
 $R_f: [0, \pi]$

Poor attempt overall



- many students drew a curve rather than straight line

- incorrect domain/range

- incorrect shape

- overall, needs more practice on sketching inverse Trigonometric functions

Q12

$$\begin{aligned} \text{a(i)} \quad A \cos(x+\alpha) \\ = A[\cos x \cos \alpha - \sin x \sin \alpha] \\ = \sqrt{3} \cos x - \sin x \quad \checkmark \end{aligned}$$

equating coefficients of $\cos x$ and $\sin x$,

$$\begin{aligned} A \cos \alpha &= \sqrt{3} \\ A \sin \alpha &= 1 \end{aligned} \quad \left. \begin{array}{l} \text{square and} \\ \text{add gives} \end{array} \right\} \quad \checkmark$$

$$A^2 = 3+1=4 \quad \checkmark$$

divide,

$$A=2 \quad (A>0)$$

$$\tan \alpha = \frac{1}{\sqrt{3}}, \quad \alpha = \pi/6; \quad 0 < \alpha < \pi/2$$

$$\therefore \sqrt{3} \cos x - \sin x = 2 \cos(x + \pi/6)$$

$$\text{(ii)} \quad 2 \cos(x + \pi/6) = 1, \quad 0 \leq x \leq 2\pi$$

$$\cos(x + \pi/6) = \frac{1}{2} \quad \frac{S+A}{T+C}$$

$$x + \pi/6 = \pi/3, \quad 2\pi - \pi/3 = 5\pi/3 \quad \checkmark$$

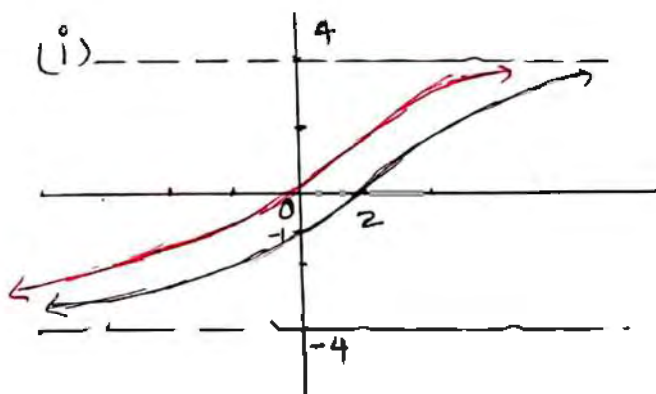
$$x = \pi/3 - \pi/6, \quad 5\pi/3 - \pi/6$$

$$x = \pi/6, \quad 3\pi/2$$

$$x = \pi/6, \quad 3\pi/2 \quad \checkmark$$

b)

$$y = f(x+2)$$



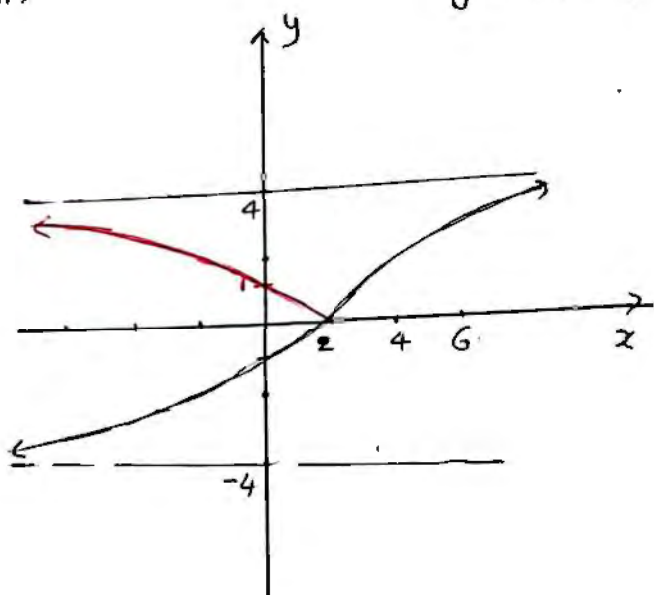
(ii)

xy

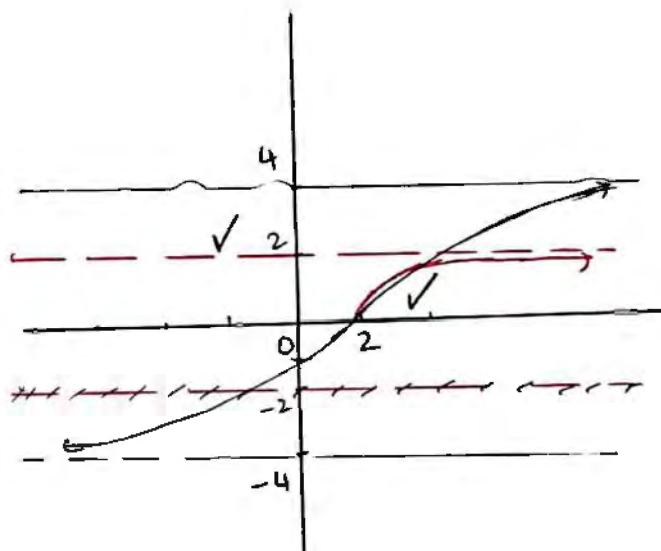
$$y = |f(x)|$$

(ii)

$$y = |f(x)|$$



(iii)



c) $\underline{a} \cdot \underline{b} = 0$ as $\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$
 $\theta = 90^\circ, \cos 90^\circ = 0$ ✓

$$3k^2 + 2(-2 - 2k) = 0$$

$$3k^2 - 4k - 4 = 0 \quad \checkmark$$

$$k = \frac{4 \pm \sqrt{16 + 4 \times 4 \times 3}}{6}$$

$$= \frac{4 \pm 8}{6} = 2, -\frac{1}{3} \quad \checkmark$$

d) $(1+x)^n - 1$ is divisible
by x for $\forall n \geq 1$.

for $n=1$,
 $(1+x)^1 - 1 = x$ which is
divisible by x .

let the result be true
for $n=k$, $k \geq 1$, $k \in \mathbb{Z}$.

$$(1+x)^k - 1 = xP, \quad P \in \mathbb{Z}$$

to prove that the result
is true for $n=k+1$,

$$\begin{aligned}(1+x)^{k+1} - 1 &= (1+x)^k \cdot (1+x) - 1 \\ &= (xP+1)(1+x) - 1 \\ &\quad \text{(using assumption)}\end{aligned}$$

$$= xP + x^2P + 1 + x - 1$$

$$= x(1 + P + xP)$$

$$= xQ, \quad Q = 1 + P + xP \in \mathbb{Z}^+$$

$\therefore$ Using PMI, $(1+x)^n - 1$

is divisible by all $n \in \mathbb{Z}$,
 $n \geq 1$, $x \in \mathbb{Z}^+$.

Question 13

(a) $4x^3 - 27x + k = 0$ has a double root

Let α, α, β be the roots

$$P'(x) = 12x^2 - 27$$

$$P'(\alpha) = 0$$

$$12\alpha^2 - 27 = 0 \quad \checkmark$$

$$\alpha^2 = \frac{27}{12} = \frac{9}{4}$$

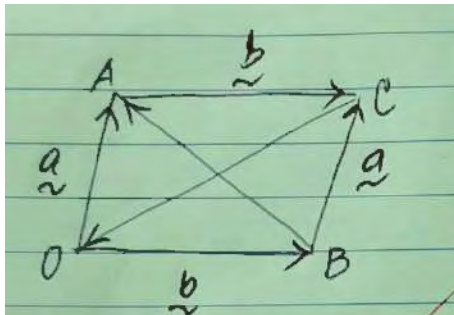
$$\alpha = \pm \frac{3}{2}$$

$$2\alpha + \beta = 0$$

$$= -2\alpha$$

$$\therefore \alpha = \frac{3}{2}, \beta = -3 \quad \checkmark$$

$$\therefore \alpha = -\frac{3}{2}, \beta = 3$$



(b)

$$\text{Let } \vec{OA} = \underline{a}, \vec{OB} = \underline{b}$$

$$\vec{OC} = \underline{a} + \underline{b}$$

$$\vec{BA} = \underline{a} - \underline{b}$$

$$|\vec{OC}|^2 + |\vec{BA}|^2$$

$$= |\underline{a} + \underline{b}|^2 + |\underline{a} - \underline{b}|^2$$

$$= (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) + (\underline{a} - \underline{b}) \cdot (\underline{a} - \underline{b})$$

$$= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b}$$

$$+ \underline{a} \cdot \underline{a} - \underline{a} \cdot \underline{b} - \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b}$$

$$= 2|\underline{a}|^2 + 2|\underline{b}|^2$$

$$= |\underline{a}|^2 + |\underline{b}|^2 + |\underline{a}|^2 + |\underline{b}|^2$$

note that parallelogram has two pairs of parallel & equal sides.

$$\alpha^2 \beta = -\frac{k}{4}$$

$$\therefore k = -4\alpha^2 \beta$$

3 marks

$$\text{For } \alpha = \frac{3}{2}, \beta = -3$$

$$k = -4\left(\frac{3}{2}\right)^2(-3)$$

$$\boxed{k = 27}$$

$$\text{For } \alpha = -\frac{3}{2}, \beta = 3$$

$$k = -4\left(-\frac{3}{2}\right)^2(3)$$

$$\boxed{k = -27}$$

OACB is a parallelogram

3 marks

Question 13

(c) $X \sim \text{Bin}(50, 0.08)$

(i) $\mu = E(X) = np$
 $= 50 \times 0.08$
 $= 4$ ✓

$\text{Var}(X) = npq$
 $= 50 \times 0.08 \times 0.92$ ✓
 $= 3.68$

$\sigma = \sqrt{3.68} = 1.92$ (2 d.p.) ✓

(ii) $P(\mu - \sigma \leq X \leq \mu + \sigma) = P(2.08 \leq X \leq 5.92)$
 $= P(X=3 \text{ or } X=4 \text{ or } X=5)$
 $= {}^{50}C_3 (0.08)^3 (0.92)^{47} + {}^{50}C_4 (0.08)^4 (0.92)^{46}$
 $+ {}^{50}C_5 (0.08)^5 (0.92)^{45}$
 $= 0.5659$ (4 d.p.)

(iii) $P(X > 5) = 1 - P(X=0 \text{ or } X=1 \text{ or } X=2 \text{ or } X=3 \text{ or } X=4 \text{ or } X=5)$
 $= 1 - [0.92^{50} + {}^{50}C_1 (0.08) (0.92)^{49} + {}^{50}C_2 (0.08)^2 (0.92)^{48}$
 $+ {}^{50}C_3 (0.08)^3 (0.92)^{47} + {}^{50}C_4 (0.08)^4 (0.92)^{46}$
 $+ {}^{50}C_5 (0.08)^5 (0.92)^{45}]$
 $= 0.2081$ (4 d.p.)

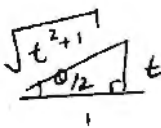
2 marks

(iv) $X \sim \text{Bin}(50, 0.04)$
 $\mu = 50 \times 0.04 = 2$
 $\sigma = \sqrt{50 \times 0.04 \times 0.96} = 1.39$
 $\mu + \sigma = 3.39$

$P(X \leq 3)$
 $= 0.96^{50} + {}^{50}C_1 (0.04) (0.96)^{49} + {}^{50}C_2 (0.04)^2 (0.96)^{48}$
 $+ {}^{50}C_3 (0.04)^3 (0.96)^{47}$
 $= 0.8609$ (4 d.p.)

2 marks

Question 14 :

a) 1) $\frac{t}{1} = \tan\left(\frac{\theta}{2}\right)$ 

$$\therefore \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$
$$= \frac{2t}{\sqrt{1+t^2}} \times \frac{1}{\sqrt{1+t^2}} = \frac{2t}{1+t^2} \quad (1)$$

$$\cos \theta = \cos^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)$$
$$= \frac{1-t^2}{1+t^2} \quad (1)$$

Done mostly well.

11) LHS = $\frac{1+\sin \theta - \cos \theta}{1+\sin \theta + \cos \theta}$

$$= \frac{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$$

$$= \frac{1+t^2+2t-1+t^2}{1+t^2+2t+1-t^2}$$

$$= \frac{2t^2+2t}{2t+2}$$

$$= \frac{2t(t+1)}{2(t+1)}$$

$$= t$$

$$= \tan \theta$$

$$= \text{RHS} \quad \therefore \text{proven}$$

Done mostly well.

CfE for errors in part 1) and some attempt was made.

$$b) \quad i) \quad \boxed{\ddot{x} = 0}$$

$$\dot{x} = \int \ddot{x} dt$$

$$= C_1$$

$$\text{at } t=0, \theta=0^\circ, \dot{x} = V \cos \theta$$

$$\therefore \dot{x} = V \cos(0) = C_1$$

$$= V$$

$$\therefore \boxed{\dot{x} = V}$$

$$x = \int \dot{x} dt = Vt + C_2$$

$$\text{at } t=0, x=0$$

$$x = V(0) + C_2 = 0$$

$$\therefore \boxed{x = Vt}$$

①

$$\boxed{\ddot{y} = -10}$$

$$\dot{y} = \int \ddot{y} dt = -10t + C_3$$

$$\text{at } t=0, \dot{y} = V \sin \theta, \theta=0^\circ$$

$$\therefore \boxed{\dot{y} = -10t}$$

$$y = \int -10 dt = -5t^2 + C_4$$

$$\text{at } t=0, y=125$$

$$C_4 = 125$$

$$\therefore \boxed{y = -5t^2 + 125}$$

①

NOTES:

- Most students showed some attempt.

- Students didn't show $\dot{x} = V \cos \theta$ and $\dot{y} = V \sin \theta$ initially, just disregarded the trig components without justification ($\theta=0^\circ$ initially)

ii) Showing some attempt at explaining displacement has horizontal and vertical component is sufficient for 1 m. ①

$$iii) \quad x = Vt$$

$$\therefore t = \frac{x}{V}$$

$$y = -5t^2 + 125$$

$$= -5\left(\frac{x}{V}\right)^2 + 125$$

①

$$\text{at } x = 20\sqrt{5} \quad y = 100 + C$$

$$100 + C = -5\left(\frac{20\sqrt{5}}{V}\right)^2 + 125$$

→

$$\Rightarrow -25 + C = \frac{-5(400 \times 5)}{V^2}$$

①

$$V^2 = \frac{10000}{25 - C}$$

$$\therefore V = \frac{100}{\sqrt{25 - C}} \quad \left(\begin{array}{l} \text{since} \\ V > 0 \end{array} \right)$$

NOTE: - Few students said $V > 0$.

- Other alternative methods were accepted.

- Not done as well.

iv) Just clears tower $\Rightarrow c=0$

$$V = \frac{100}{\sqrt{25-0}} \quad (1)$$

$$= \frac{100}{5}$$

$$= 20 \text{ m/s as required.}$$

NOTE:

Done well

v) Horizontal range $\Rightarrow y=0, \quad V=20 \text{ m/s}$

$$y = \frac{-5x^2}{20^2} + 125 = 0$$

$$x^2 = 10000$$

$$x = 100 \text{ m (since } x > 0) \quad (1)$$

NOTE:

- Some students misunderstood this while others got CFE for making conclusion on how far past tower.

$$\therefore 100 - 20\sqrt{5} \text{ m past the tower} \quad (1)$$

vi) $y=0$

$$\Rightarrow 0 = 5t^2 - 125$$

$$125 = 5t^2$$

$$t = 5 \text{ (} t > 0 \text{)} \quad (1)$$

$$\therefore \tan \theta = \frac{\dot{y}}{\dot{x}} = \frac{-10(5)}{20} = \frac{-50}{20}$$

$$\theta = \tan^{-1}\left(\frac{-50}{20}\right)$$

$$\approx 112^\circ \text{ or } 68^\circ \text{ from the ground.} \quad (1)$$

$$v = \sqrt{(-50)^2 + 20^2}$$

$$= 10\sqrt{29} \text{ m/s. impact velocity.} \quad (1)$$